

2) Leetcode 8316

constraints :
 • $1 \leq s.length \leq 10^4$
 • s consists of lowercase English letters.

```
char* removeDuplicateLetters(char* s){
```

```
    int n = strlen(s);
```

```
    int count[26] = {0};
```

```
    bool inStack[26] = {false};
```

```
    for (int i = 0; i < n; i++){
```

```
        count[s[i] - 'a']++;
```

```
    }
```

```
    char *stack = (char*) malloc((n+1)*sizeof(char);
```

```
    int top = 0;
```

```
    for (int i = 0; i < n; i++){
```

```
        char c = s[i];
```

```
        count[c - 'a']--;
```

```
        if (inStack[c - 'a']) continue;
```

```
        while (top > 0 && stack[top-1] > c &&
```

```
            count[stack[top-1] - 'a'] > 0){
```

```
            inStack[stack[top-1] - 'a'] = false;
```

```
            top--;
```

```
    }
```

```
stack[top++] = c;  
inStack[c - 'a'] = true;  
}
```

```
stack[top] = '\0';
```

```
return stack;
```

```
}
```

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